Helix Robotics: Helios Autonomous Mobile Robot - Technical Brief

Helix Robotics designs and manufactures autonomous mobile robots for warehouse and fulfillment operations. The company was founded in 2018 by David Chen and is headquartered in Toronto, Canada. Helix's chief technology officer is Priya Sharma, who joined the company in 2019 from Boston Dynamics, where she led the perception team for the Stretch robot. Helix operates a 40,000 square foot manufacturing facility in Mississauga, Ontario, and expects to double production capacity by the end of 2026.

The company's flagship product is the Helios autonomous mobile robot. Helios is rated to carry a payload of 25 kilograms and operates for up to 8 hours on a single battery charge. The robot navigates using a combination of LiDAR and a 3D computer vision stack, reaching a top speed of 3 meters per second in corridors. Helios is designed for high-density pallet handling and can swap batteries without halting a fulfillment shift. Each unit is covered by a five-year service contract that includes predictive maintenance through Helix's cloud fleet-management portal.

Helix's largest customer is Nova Dynamics, which began deploying Helios units in 2023 under a partnership agreement and operated more than 1,200 units across its North American fulfillment centers by the end of 2025. Northwind Logistics, a freight and fulfillment operator based in Atlanta, placed a 300-unit order in 2025, with deliveries scheduled through 2026. Helix also runs a joint research and development facility with Nova Dynamics in Toronto focused on autonomous retrieval systems for high-value inventory.

On the roadmap, Helix plans to launch Helios 2 in the fourth quarter of 2026 with a 40 kilogram payload capacity and extended eight-and-a-half-hour runtime. The company expects to ship 5,000 additional Helios units in 2026 across its existing customer base. Helix continues to evaluate entry into grocery e-commerce fulfilment, though management has not committed to a timeline for that market.